RD900 locus structure project description

Background

We have previously published a paper detailing variation across the Mycobacterium tuberculosis Complex (MTBC) in the RD900 region. This region was first described as a lineage specific locus in M. africanum GM041182 strain (Bentley et al. 2012) and was thought to be deleted in M. bovis and "modern" M. tuberculosis lineages. RD900 region was not present in the original M. bovis AF2122/97 reference genome annotation, but was found to be actually present upon re-sequencing. These species show more than 99% genetic identity but exhibit distinct host preference and virulence. Differences in key locii are likely to be important in determining this.

The RD900 region comprises two homologous genes, pknH1 and pknH2, encoding a serine/threonine protein kinase PknH flanking the tbd2 gene. Our analysis revealed that RD900 has been independently lost in different MTBC lineages and different strains, resulting in the generation of a single pknH gene. Importantly, all the analysed M. bovis and M. caprae strains carry a conserved deletion within a proline rich-region of pknH, independent of the presence or absence of RD900. We hypothesized that deletion of pknH proline rich-region in M. bovis may affect PknH function, having a potential role in its virulence and evolutionary adaptation.

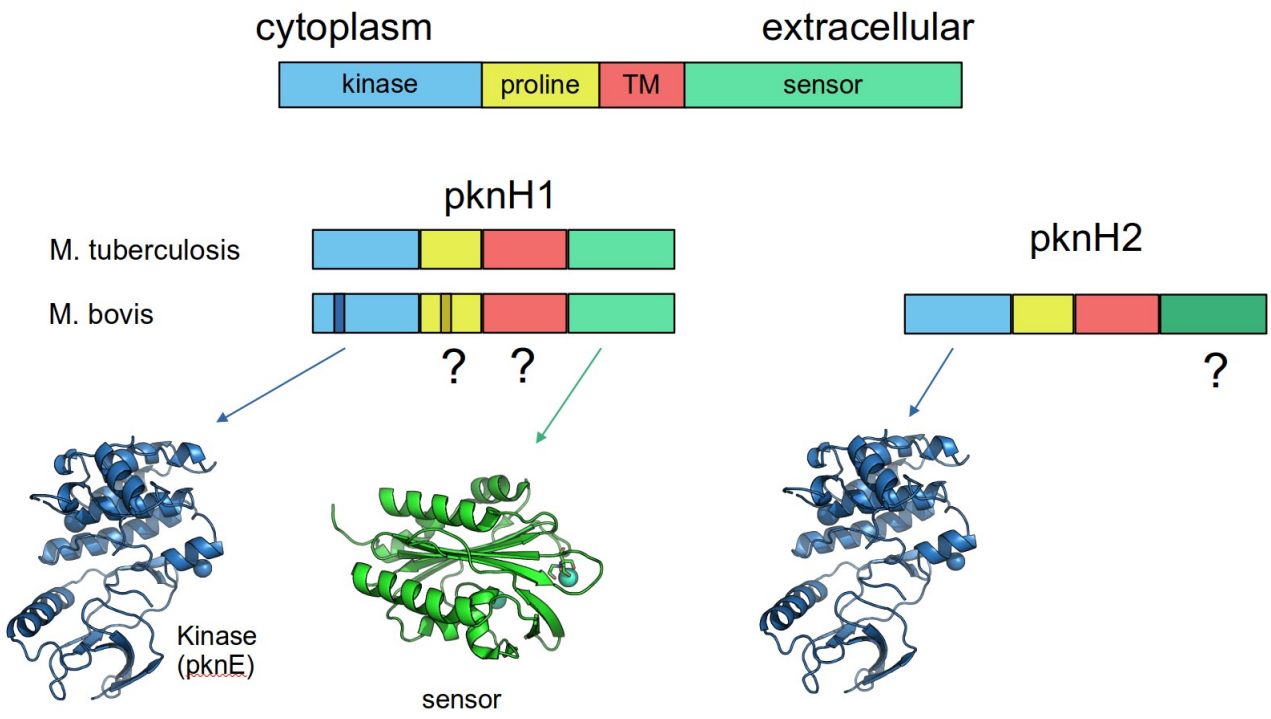


Figure 1: Structure of the pknH1 and pknH2 genes.

In this study we examined over 60 different MTBC strains representing lineages across the MTBC to see variations in the region. To accurately detect locus structure we used the raw read data instead of

relying on assembled genomes which may have errors due to homology in the kinase domains of the pknH proteins, if both are present. Raw fastq files for all strains were downloaded and aligned to the most complete version of the RD900 locus encoded in the Mycobacterium africanum (MAF) genome. This allowed missing features to be seen as gaps, as shown below.

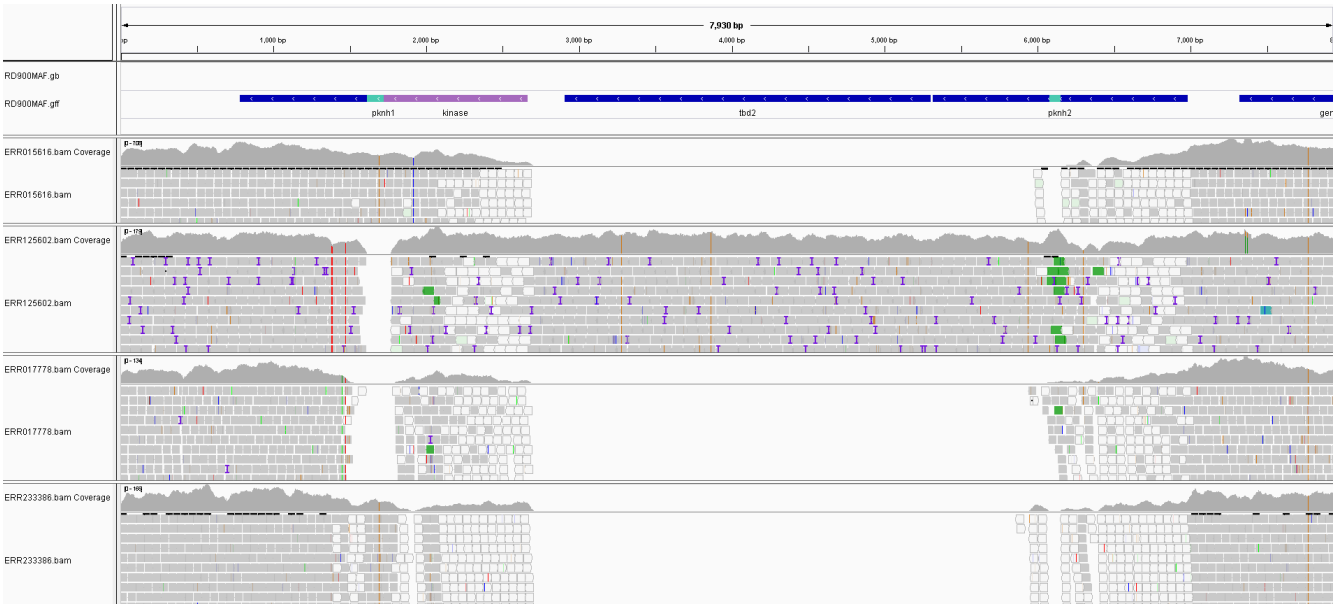


Figure 2: Alignments of some samples to the RD900 MAF region show the gaps where features are missing. The white reads are those multi-mapping to the homologous kinase domains of pknH1. This is what causes issues in assembly at this locus.

Objectives

Since this project was done in 2019, new data has been produced for all of these mycobacteria genomes including the animal adapted species we are most interested in. We will use this new data to provide a more complete analysis. As well as using raw read data as before we will use complete assemblies as a complementary approach. Complete assemblies should cover the whole locus and give a reliable picture of it's structure.

The question

RD900 is a ~3.1 kb region of the *Mycobacterium tuberculosis* complex (MTBC) genome. It contains the duplicated kinase genes *pknH1* and *pknH2*, the gene *tbd2*, and flanking genes. It is intact in *M. africanum* GM041182 but has been deleted independently in different lineages, including modern *M. tuberculosis* and the *M. bovis* lineage. Some strains carry partial deletions. We want to know, for each sample, which parts of this region are present.

We will:

- Retrieve new raw read post 2019 data to apply the same approach as before

- We will find all complete assemblies for the same set of species. In particular this may help find a more complete locus for Africanum. which we can align to.
- In both cases we need to reduce redundancy drastically to get a manageable dataset. I will handle this part.
- It might be that we can perform both methods in parallel and then compare results but one will feed into the other.

Methods

1. The assembly approach

There are >600 compete MTBC assemblies and >120 incomplete long read assemblies available. These have already been downloaded. We actually do not need to reduce these now, as we can just extract the locus for all of them first and then reduce redundancy at the locus level.

Extracting the RD900 locus

We chose to adopt an anchor based extraction + alignment method. We find the conserved anchor areas around the locus first, then **BLAST the two flanking anchor genes separately** (not the internal content), then take whatever lies between their hit coordinates in each assembly as "the locus," regardless of what's in there. This sidesteps the paralog-confusion problem and should correctly extract the interval *whatever its content*: intact, deleted, partial, or bearing a large insertion, with no assumption about what should be inside.

Comments/caveats on this approach in general:

- Anchor choice matters more than anything else. They need to be genuinely unique and conserved across *all* species in - so it's worth explicitly checking copy number of your chosen anchor genes across the assemblies.
- Assembly fragmentation breaks it silently. But we are using complete genomes so that should be ok, but do not assume 100% extraction across all assemblies.

Annotate-cluster-reduce method for grouping locus sequences

Since we want to see all possible diversity in the locus across species but avoid duplication, we want to do the following:

1. annotate the protein structure for all sequences using our known proteins
2. cluster/group by structure
3. mash reduce within each group and get a representative sequence

We will also be interested in whether the same species is in >1 structure groups (should be the case for *M.bovis*).

On threshold: 0.001 is a good starting point for collapsing near-duplicate locus sequences. But use different thresholds to see the results. If we use very small thresholds we will get too many groups.

Find lineages

This will be done with a tool called TB-profiler. I will do this part and add it to the table.

Compare the RD900 sequences

We can do a multiple sequence alignment of the locus for all samples. Using mafft is fastest.
We can also build a phylogeny with *fasttree* to get an idea of how the locus sequences line up as compared to their species relationship.
To compare protein structure of the locus groups we can do it visually with Clinker or Python (see below).

Analysis steps

Note: when using Python scripts you can always run python <scriptname.py> -h to get help.

Here are some of the commands for the assembly analysis. Your filenames may differ in some cases.
Extract the up and downstream anchors:
Here is the link to the genome viewer for the reference Africanum genome where you can extract the flanking sequences from the region.
https://www.ncbi.nlm.nih.gov/nuccore/FR878060.1?report=graph&from=1411192&to=1416081

The script locus_extraction.py will then extract the locus sequences. Note that you can run the extraction on the complete and incomplete genomes separately and then combine the resulting fasta together (using cat). Or put all genomes in one folder if you prefer.

```
python locus_extraction.py --up upstream_anchor.fa --down downstream_anchor.fa --assemblies mtb_genomes
```

The script annotate.py will identify all the proteins in each locus and create a gbk file for each one in a folder.

```
python annotate.py -p RD900MAF.faa -o annot RD900_extracted.fa
```

To cluster the locus sequences after annotation we use this script:

```
python locus_cluster.py --input RD900_extracted.fa --gbk annot --threshold 0.0001
```

This will produce an output folder structure_results where there is also a structure summary table. This will be useful for curating the sequences we want to select ourselves if needed. It also writes out the representative gbk files to a folder called annot_repr and the sequences to a fasta file inside structure_results.

We can use clinker to view the output

```
clinker annot/*.gbk -o clinker_alignment.html -p clinker_plot.html -i 0.75
```

There is also a custom Python script to plot the locus comparison that will be more easy to customise later. This will plot whatever gbk files you give it. So you can only give it the representative ones from the structure results table for example.

```
python locus_plot.py --input RD900_extracted.fa --gbk annot --table asm_complete_mtbc.csv
```

Sequence alignment

Align the representative sequences:

```
mafft --auto --thread 4 RD900_extracted.fa > RD900_extracted_repr.fa
```

Make a phylogeny from the alignment:

```
fasttree -nt -gtr RD900_extracted_aligned.fa > RD900_locus_tree.nwk
```

Further analysis and interepretation

We want to establish the canonical locus for each species especially were it is split into multiple types. Outliers and singletons are most suspect (but could be real). Any of the genomes can be checked first for the locus sequence region on the genbank graphical viewer. We can see how the locus is annotated there. Is it different? Possibly we can use another annotation tool to double check specific results.
You can use the structure_summary file to curate the representative gbk files into a separate folder for plotting the final set. First add the ones we are sure of.

Things to look for:

- There may be unusual outliers and they may be incorrect. Caused by wrong annotation? Are they real?
- Are the results we see consistent with the previous study? Do we have new findings?
- Does the same species appear in multiple structure groups?
- Is there a larger version of the M. Africanum locus than the one that we are currently using as a reference? Probably not but worth checking.

Notes:

One example is mTB the n=29 pknH1* - pknH1* split is evidence for real biology, not against it. (number in each subcluster is threshold dependent).

Look past the raw count to the n_subclusters column: n members split across independent Mash subclusters. Since our Mash sub-clustering operates within a structural group on sequence similarity, 7 subclusters means this exact disruption pattern (same two-fragment split, same gene order) arose in at least 7 genetically distinct backgrounds - not one clonal outbreak that got oversampled. An extraction or annotation artifact tied to one bad assembly, one bad batch, or one submitter's pipeline would be expected to behave idiosyncratically per genome, not reproduce an identical structural signature across unrelated lineages. Independent recurrence across divergent backgrounds is close to a self-replicating experiment.

It's also exactly what the literature predicts: We found RD900-region genes are lost/disrupted recurrently and homoplasically across the MTBC tree, not as a single clade-defining event. A frameshift that keeps reappearing independently is the expected signature of that, maybe not an anomaly.

One thing worth checking before we fully believe it, though: "pknH1* - pknH1*" in our structure string doesn't by itself distinguish a true frameshift from two genuinely separate partial hits. The overlap-resolution logic only keeps two same-gene fragments together when they overlap in genomic coordinates (the frameshift signature — same DNA, different frame). Two non-overlapping fragments for the same gene can also pass through if there's a gap between them with no hit at all — which would

point at something else (a real internal deletion, or a repeat/low-complexity stretch inside pknH1 defeating the lightweight k-mer aligner) rather than a clean frameshift. Worth pulling the actual start/end coordinates for a handful of these 29 and checking: (a) do the two fragments overlap or sit end-to-end with a gap, and (b) does the break point land at the same codon/position across most of the 29 — which would point to a specific mutational hotspot (classically a homopolymer or short tandem repeat prone to slipped-strand mispairing) rather than scattered independent mutations. Either answer is interesting; I can write that diagnostic if useful. I'd also run classify_gene_region() on these 29 directly — the annotation script's frameshift_candidate tag is a lighter-weight signal than the dedicated tblastn HSP-pattern check, and this is exactly the case worth spending the heavier, independent confirmation on.

Singletons

The risk profile here is different from the n=29 case, and it's specifically lack of corroboration, not rarity. A structural call seen in only one genome has nothing else to compare against, so there's no way to rule out it's a single assembly/sequencing artifact in that one sample — a homopolymer miscall (a known failure mode for some ONT chemistries, and not absent from Illumina either) can manufacture a fake frameshift that looks identical in your output to a real one.

We can report singletons separately in whatever summary we write up - e.g. "structural types confirmed in ≥2 independent genomes" vs "private/singleton variants, individually DNA-confirmed" - rather than merging or silently cutting them. Dropping rare-but-real variants would undercut the whole point of this validation pass.

2. Protein based analysis

This approach allows us to compare the protein sequences for specific proteins where they are present, especially pknH1 and 2.
We can extract the amino acid sequences and align them to see where the most variation is and where there are deletions/insertions. The amino acid sequences are already in the genbank annotations we made. Use mafft or similar to align them.
We may also predict the structure of these proteins and see if there is any predictable effects on structure.

Analysis steps

Write a Python script to extract all the sequences for a specific protein. Try pknH1 first but it should work for any of them.
Reduce the sequences as before with mash.
Align the proteins sequences and plot them.

3. Raw reads alignment approach

This is the method we used before. It avoids the issues of erroneous assembly which may exist even in the complete ones. Especially in regions like RD900. Also there is a larger raw read dataset (as not all raw WGS data is assembled and uploaded and most are never made into complete genomes).
Duplicated/paralogous content is where assemblers choke. RD900 contains two paralogous genes (pknH1, pknH2) in close proximity.

Why we use raw reads rather than assemblies

Draft assemblies are often broken into pieces (contigs) at repeated or duplicated sequence, and *pknH1* and *pknH2* are similar enough to cause this. Short-read assemblers (SPAdes, etc.) struggle precisely with repeat/paralog regions like this — reads multi-map across the two copies, and the assembler often collapses them into one contig, breaks the contig at the repeat boundary, or produces a chimeric/misassembled junction. A contig break here doesn't mean the region is deleted; it means the assembler gave up, and you can't distinguish "real deletion" from "assembly artifact" from the assembly alone. Raw reads avoid this because they are the data before any assembly decisions were made.

We want to analyze deletion patterns at a target genomic locus (including flanking sequences with the embR, AbmR, pknH1, and pknH2 genes) using sequencing reads (short reads and/or long reads like Nanopore/PacBio).

Simple Method

This approach directly targets the region of interest for alignment only. How it works: Align all samples directly to the locus sequence (including some flanking sequence). It is straightforward and simple to implement. However, it risks having multi-mappers—reads that actually originate from other parts of the genome—aligning to the locus, which creates background noise.

Steps:

1. Align all samples to the locus only sequence to produce a BAM file for each sample.
2. Count depth in 100 bp bins
3. Try to detect missing regions with an average base depth threshold in the bin and mark them as absent or present. Vary thresholds.
4. Summarise across samples: Build a samples matrix, cluster it as a heatmap with colour bars (for species, study, and sequencing platform). Contiguous low-coverage bins are turned into deletion intervals, grouping samples with the same pattern so hundreds of samples reduce to a handful of patterns (requiring IGV inspection of only one representative per group plus any rare ones).

Complex Method: Full-Genome Analysis & Advanced Processing

Note: This method may be used if higher accuracy and noise reduction are required. Some of the details here could also be applied to refine the simple method.
Instead of isolating the locus, this method maps reads to the complete reference genome (M. africanum GM041182 using BWA-MEM for short reads and minimap2 for Nanopore/PacBio). A read lands where its sequence best matches the reference; if a sample has lost a stretch of DNA, no reads will align there.

Steps:

- Count depth in 100 bp bins: Split the genome into 100 bp windows and calculate the mean read depth in each (using mosdepth), keeping the locus plus about 3 kb of flanking sequence on each side. Fixed bins are unbiased because they pick up any deletion (even those with unexpected breakpoints) without assuming where it starts and ends. Bins are then labeled with gene names for gene-by-gene review.
- Normalise to each sample's own baseline: Raw depth differs between samples (e.g., 150x vs. 30x). Divide each bin's depth by that sample's median depth across the rest of the genome (median is used because it is resilient to repeats/duplications).
    - *Scale:* 1.0 = normal single-copy coverage, 0 = absent, values near 0.5 = half coverage.
- Interpret the ratios: Because genes like pknH1 and pknH2 are similar, reads from one copy can map to either reference copy.
    - If one copy is deleted, the remaining copy's reads are shared between both reference positions (showing ~0.5x coverage rather than 0).
    - Bases unique to a single copy drop to near zero if that copy is lost.
- Apply display thresholds: Keep ratios as continuous values during analysis, simplifying them only for display: below ~0.2 (absent), ~0.2–0.8 (reduced), above ~0.8 (present). *(Note: These thresholds are starting points and should be verified against real data).*
- Summarise across samples: Same as above

Why Align to the Whole Genome Rather Than Just the Locus?

- **Fewer false signals:** Mapping to an isolated 3 kb locus forces similar sequences from elsewhere in the genome (such as PE/PPE gene families) to pile onto the locus, making it look present when it isn't. Whole-genome mapping sends those reads to their true locations.
- **A built-in baseline:** Genome-wide depth is necessary to normalise each sample; a locus-only reference gives nothing to compare against.
- **Unique flanking regions as controls:** If the flanks look normal and the middle is empty, you can be confident the region is genuinely missing rather than just poorly sequenced.
- **Consistency:** Every sample is processed identically, ensuring differences reflect biology rather than handling discrepancies.
- Limitations to Keep in Mind:
- **Low-coverage samples:** Samples below ~20x give noisy bins and should be flagged for exclusion.
- **Divergent lineages:** Can show coverage dips where there are many SNPs relative to the reference (a shallow dip is not necessarily a deletion).
- **Mixed infections:** Produce intermediate coverage ratios.
- **Repeats and paralogs:** These regions are sensitive to mapping-quality filters. Run the analysis both with and without the filter to compare.
- **Long reads validation:** Nanopore/PacBio long reads can span the whole locus in a single read, providing a direct check on the coverage-based results.

Datasets

On the shared drive there are now several sets of data:

mtbc_assemblies.zip: this zip file contains all ~680 complete assemblies.
mtbc_assemblies_lr.zip: this zip file contains incomplete long read assemblies.
asm_all_mtbc.csv: the list of details for all the assemblies. Includes the country. This will be useful later for identification.

Software installation

On Ubuntu for windows (WSL) you can install most software with apt. e.g. sudo apt install bwa. Unfortunately a small number of bioinformatics tools, even those written in Python, are much easier to install with conda. Conda is a last resort if we can't install another way.
On macOS it may be installable with brew, if not conda may be needed.